ESILV - Tutorial #3

Commodity Markets and Models

- This session (worth **5%**) lasts only 1h30.
- Solve exercises **on paper** and hand it in to your tutorial leader (*one per group*) before the end. (no email) – (no late submission)
- Ask your instructor for advice. Corrections will not be posted on DVL.
- In case of absence to this tutorial, you will be graded as zero. **Your instructor does not need to know the reason of your absence.** Official absence needs to be recorded directly with the scholar office (scolarite-esilv@devinci.fr).

Problem 1: `State Space Models for Soybean Price Analysis`

In commodity markets, particularly for storable agricultural commodities like soybeans, inventory levels are fundamental to price volatility and forward curve dynamics. Geman and Nguyen (2005) demonstrated that soybean price volatility exhibits a linear relationship with *scarcity*.

- Geman, H., & Nguyen, V.N. (2005). Soybean Inventory and Forward Curve Dynamics. *Management Science*, 51(7), 1076–1091.

Introduction: The Theory of Storage

In agricultural markets, the price volatility (σ) is not constant. It depends on Scarcity. Let I_t be the level of soybean inventory. We define the Scarcity Index s_t as:

$$s_t = \frac{1}{I_t} \tag{1}$$

Empirical evidence suggests that when inventories are low, price volatility spikes.

Question #1: Economic Intuition

a) Based on the formula $s_t = 1/I_t$, what happens to volatility when inventory approaches zero? Explain why a "stockout" makes prices more sensitive to small supply shocks.

b) In a "Contango" market (Forward Price > Spot Price), do you expect inventories to be high or low? Why?

State Space Representation

To track the "hidden" state of the market, we use a simplified two-factor model where the Log-Price (y_t) depends on a long-term trend (M_t) and a short-term deviation (B_t).
The system is defined by:

- **Transition:** $M_{t+1} = M_t + B_t + \eta_t$
- **Trend:** $B_{t+1} = B_t + \zeta_t$

- **Observation:** $y_t = M_t + \epsilon_t$

Question #2: Matrix Setup

Rewrite this system in the standard State-Space matrix form. *Hint: Your state vector is* $X_t = [M_t, B_t]^T$.

State Equation:

$$\begin{bmatrix} M_{t+1} \\ B_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} M_t \\ B_t \end{bmatrix} + \begin{bmatrix} \eta_t \\ \zeta_t \end{bmatrix}$$

Observation Equation:

$$y_t = \underbrace{\begin{bmatrix} ? & ? \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} M_t \\ B_t \end{bmatrix} + \epsilon_t$$

Three-State Model for Soybeans

The three-state-variable model extends the framework to include stochastic volatility:

$$\ln S_t = h_t + X_t \tag{2}$$

$$h_t = \eta \sin(2\pi t + \phi) + \theta \tag{3}$$

where S_t is the spot price, h_t captures seasonality, and X_t is the deseasonalized component. Under the risk-neutral measure $\mathbb{Q}$, the state dynamics are:

$$dX_t = \kappa(v_{1t} - X_t)dt + e^{\psi_t}\sqrt{v_{2t}}dW_t^X \tag{4}$$

$$dv_{1t} = a_1(b_1 - v_{1t})dt + \sigma_1 e^{\psi_t}\sqrt{v_{2t}}dW_t^{v_1} \tag{5}$$

$$dv_{2t} = a_2(b_2 - v_{2t})dt + \sigma_2\sqrt{v_{2t}}dW_t^{v_2} \tag{6}$$

where:

- X_t: deseasonalized log spot price
- v_{1t}: short-term mean of X_t
- $v_{2t} = (\alpha + \beta s_t)^2$: variance (scarcity-dependent)
- $\psi_t = \zeta \sin(2\pi t + \xi)$: seasonal volatility adjustment
- $s_t = 1/I_t$: scarcity (inverse inventory)

The log-likelihood function for Gaussian state space models is:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{nw}{2}\log(2\pi) - \frac{1}{2}\sum_{j=1}^{n}\log|\Delta_j| - \frac{1}{2}\sum_{j=1}^{n} I_j'\Delta_j^{-1}I_j \tag{7}$$

where $I_t = Y_t - \mathbf{G}_t\hat{\mathbf{X}}_{t|t-1}$ are the innovations and Δ_t are their variances.

Question #3: Analyze Tables

a) Explain the economic interpretation of each state variable and why mean-reversion is appropriate for commodity prices.

b) Discuss why v_{2t} is linked to scarcity through the relationship $v_{2t} = (\alpha + \beta s_t)^2$.

c) For the three-state soybean model, the parameter vector is:

$$\boldsymbol{\theta} = (\kappa, \sigma, a_1, b_1, \sigma_1, a_2, b_2, \sigma_2, \alpha, \beta, \rho_{13}, \eta, \phi, \theta, \zeta)$$

Describe the numerical optimization procedure to obtain $\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$.

d) What constraints should be imposed during optimization to ensure economically meaningful parameters?

e) Based on the parameter estimates in Table 1, discuss:

- Mean reversion speeds (κ, a_1, a_2)
- The scarcity effect ($\beta \approx 8127$)
- Correlation structure (particularly ρ_{13})

f) Interpret Table 2. Does the three-state model improve out-of-sample forecasting?

Empirical Analysis

Refer to the provided Figure 1 (Soybean Prices vs. Inventory).

Question #4: Visual Interpretation

Observe the price spikes in the data (e.g., around historical drought periods).

a) Identify a period where Inventory (I_t) was at a local minimum. What happened to the Spot Price at that exact moment?

b) The "Samuelson Effect" states that volatility is higher for contracts nearing delivery (Spot) than for long-dated contracts. Looking at the relationship between inventory and price, why does this make sense for a harvest-based commodity like soybeans?

c) How would a soybean producer use this model for hedging?

Conclusion

Why is the Kalman Filter useful for a commodity trader even if they don't observe the "true" inventory levels in real-time?

End of Exercise. Please hand in your exercise on paper to your instructor.

Table 1: Parameter Estimates from State Space Models

Parameter	Two-Factor	Three-Factor
κ (mean reversion)	0.6821 (0.0052)	0.5931 (0.0049)
σ	0.2310 (0.0002)	–
σ_1	–	2.2410 (0.0975)
a_1	1.7927 (0.0236)	1.9327 (0.0224)
b_1	8.7600 (0.0025)	8.8600 (0.0023)
ξ_1	0.4981 (0.0112)	–
a_2	–	1.1529 (0.0440)
b_2	–	0.0128 (0.0002)
σ_2	–	5.5112 (1.8347)
ρ_{13}	–	0.9585 (0.2597)
η	0.0195 (0.0001)	0.0210 (0.0001)
ϕ	−0.1025 (0.0004)	−0.0125 (0.0004)
θ	0.5575 (0.0023)	0.7575 (0.0021)
ζ	−0.0715 (0.0002)	−0.1015 (0.0002)
α	–	0.0025 (0.0009)
β	–	8127 (129)

Note: Standard errors in parentheses.

Table 2: Prediction Errors: Two-State vs. Three-State Models (2000–2026)

Contract	Two-State Model		Three-State Model	
	RMSE	Error %	RMSE	Error %
F1	0.0267	0.2618	0.0260	0.2327
F2	0.0362	0.3013	0.0338	0.2795
F3	0.0271	0.0014	0.0253	0.0920
F4	0.0226	0.1373	0.0269	0.3489
F5	0.0372	0.3725	0.0289	0.3006
F6	0.0265	0.1014	0.0241	0.1431
F7	0.0321	0.2041	0.0243	0.1512
F8	0.0263	0.0666	0.0277	0.1772
F9	0.0295	0.3539	0.0166	0.0532
F10	0.0280	0.2451	0.0220	0.0617
F11	0.0246	0.3045	0.0196	0.0976
F12	0.0233	0.2140	0.0191	0.2339
In-sample (F1–F5)	0.0299	0.2149	0.0282	0.2507
Out-of-sample (F6–F12)	0.0272	0.2128	0.0219	0.1311

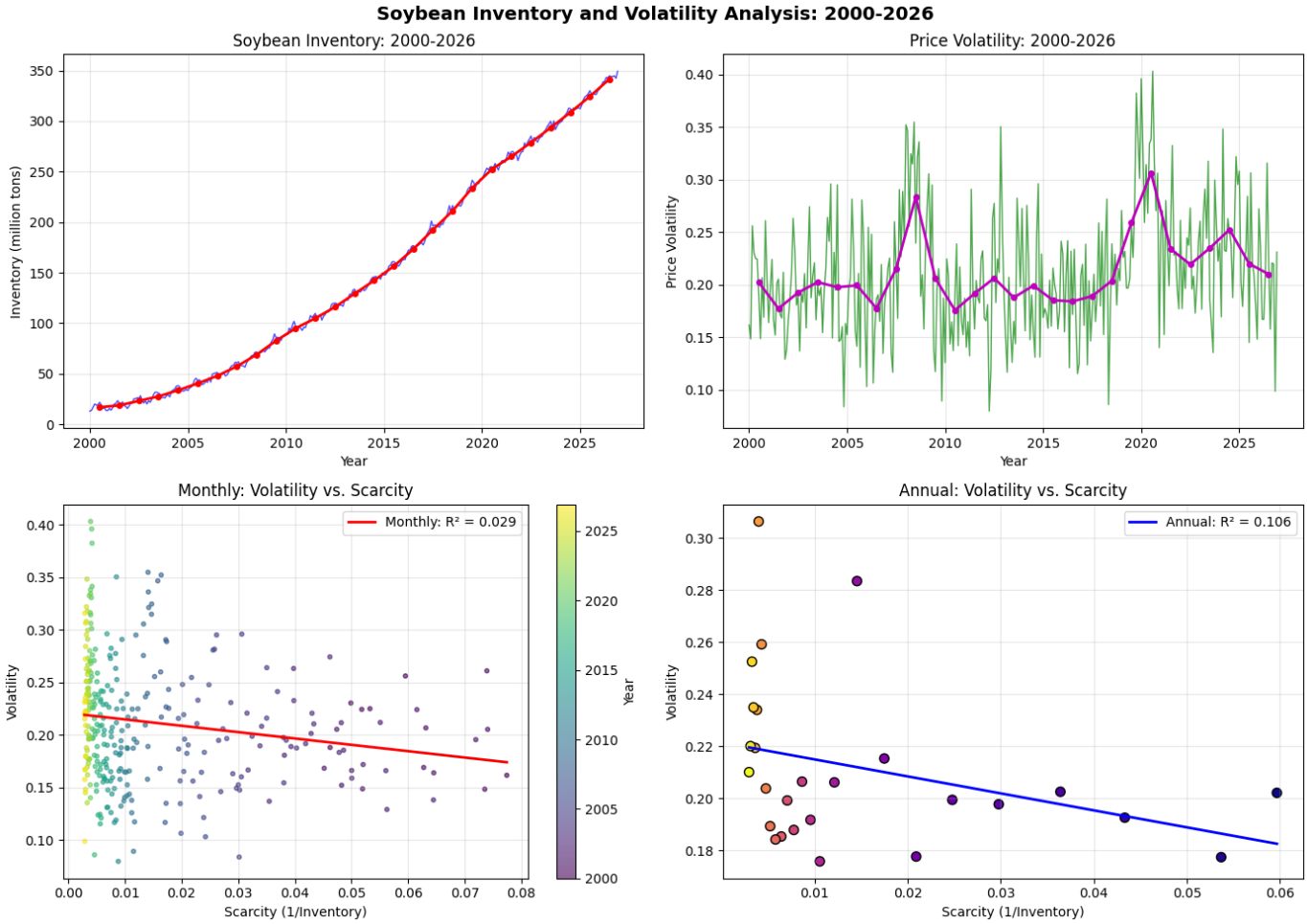

Figure 1: (*from top to bottom, left to right)* Inventory time series, Volatility time series, Volatility vs Scarcity (monthly), Volatility vs Scarcity (annual)

Table 3: Regression Summary and Descriptive Statistics

Description	**Value**
Monthly Data	
Intercept	0.2208
Slope	-0.6039
R-squared	0.0286
P-value	0.002262
Annual Data	
Intercept	0.2214
Slope	-0.6525
R-squared	0.1064
P-value	0.096816
Descriptive Statistics	
Monthly inventory mean	149.44 million tons
Monthly inventory std	104.61
Monthly volatility mean	0.212
Monthly volatility std	0.059